

Technical Specification v0.9

Frontend Architecture & UX Specification

Já — þá höldum við áfram með sama architecture-mynstri. Í v0.9 myndi ég sérstaklega passa að frontend-ið verði **case-centric**, þannig að lögfræðingurinn vinni að mestu úr einu samræmdu *Case Workspace* í stað þess að hoppa á milli óskyldra skjáa.

Technical Specification v0.9

Frontend Architecture & UX Specification

Project: Icelandic Legal Case Management System — ILCMS

Version: 0.9

Status: Frontend Architecture Baseline

Date: September 2026

1. Purpose

Technical Specification v0.9 skilgreinir frontend architecture og UX fyrir ILCMS.

Markmiðið er að skapa:

- case-centric user experience
- consistent navigation
- role-aware UI
- secure document handling
- efficient legal workflows
- audit visibility
- accessible interface
- responsive desktop-first application
- reusable frontend architecture

Frontend skal ekki einfaldlega endurspegla REST API-ið.

Það skal endurspegla **legal workflow** og **user mental model**.

2. Technology Baseline

Recommended stack:

React
TypeScript
Vite
React Router
TanStack Query
Zustand
React Hook Form
Zod
UI:

Component library
Design system
CSS / utility framework
Exact UI library er intentionally ekki locked í þessari útgáfu.

3. Browser Support

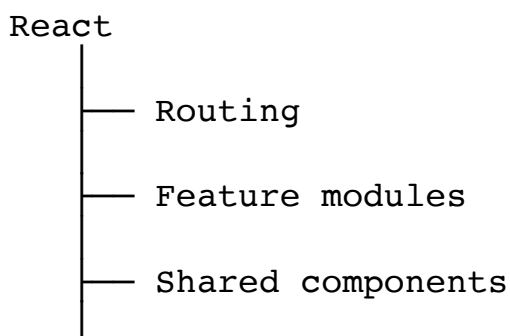
Supported browsers:

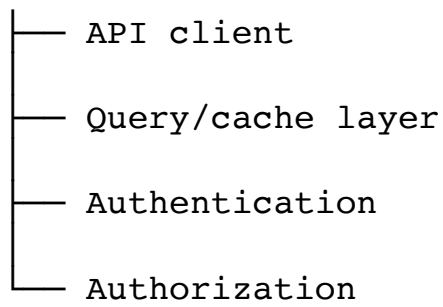
Chrome
Edge
Firefox
Safari
Priority:

Desktop
Laptop
Tablet
Mobile phone er ekki primary target fyrir fulla case management experience.

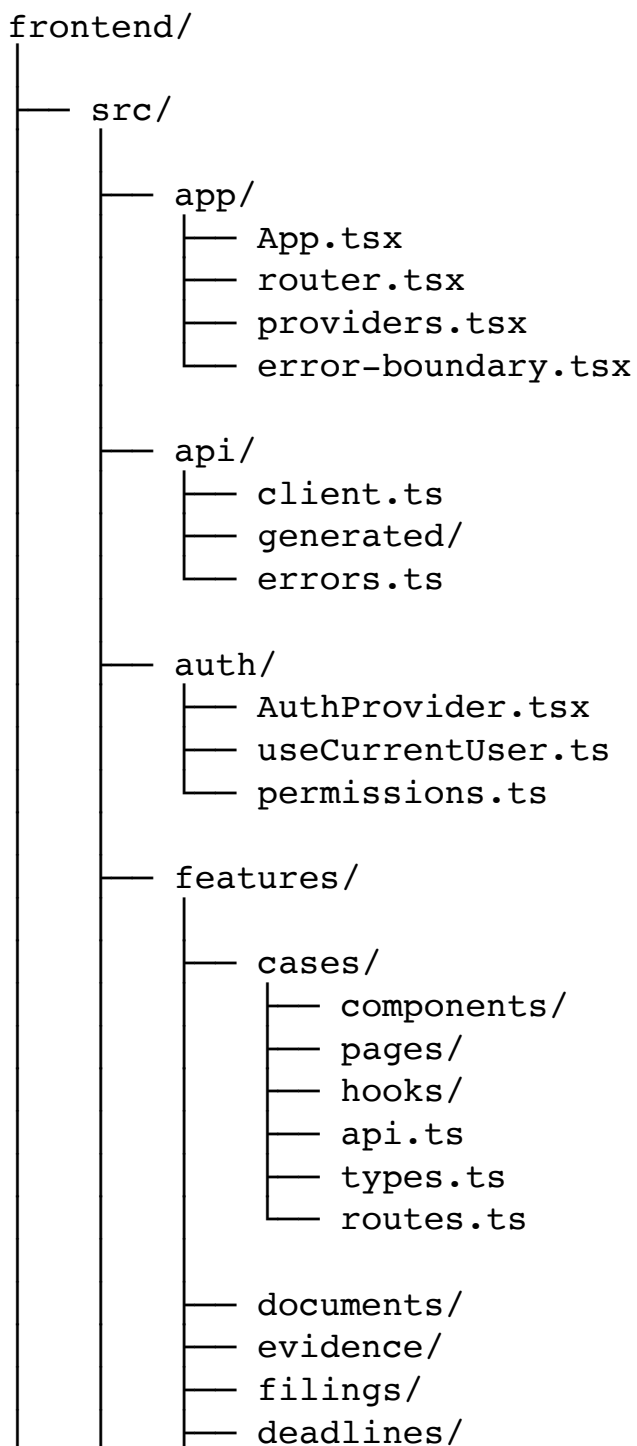
4. Frontend Architecture

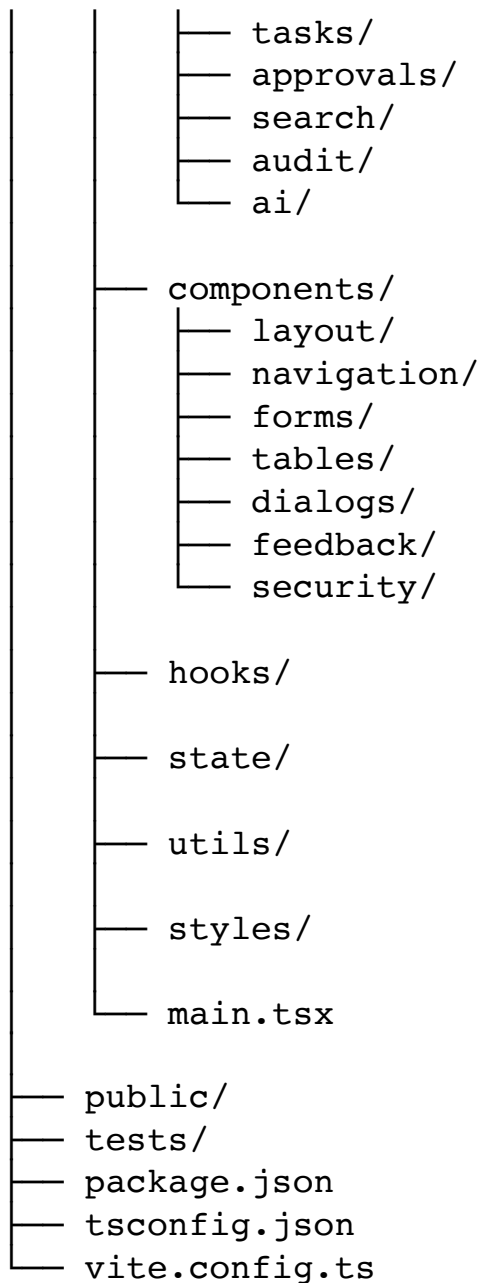
Recommended:





5. Project Structure





6. Feature-Based Architecture

Feature modules skulu vera sem mest sjálfstæðar.

Dæmi:

```
features/cases/  
features/documents/  
features/evidence/
```

Ekki:

```
components/  
  caseComponent.tsx  
  documentComponent.tsx  
  evidenceComponent.tsx
```

með allri business logic þar.

7. Design Principle

UI components:

presentation

Hooks:

frontend behavior

API:

server communication

Backend:

business authority

Frontend má aldrei vera authoritative fyrir security eða legal business rules.

8. Application Shell

Basic application:

ILCMS	Search	User ▼
Dashboard Cases Documents Evidence Filings Deadlines Tasks Search Admin	Main Content	

Navigation er role-aware.

9. Global Navigation

Primary navigation:

Dashboard
Cases
Documents
Evidence
Filings
Deadlines
Tasks
Search
Administrative users sjá einnig:

Administration
Users
Roles
Organizations
Retention
Audit
System

10. Case-Centric Navigation

Pegar notandi opnar case:

- Case
- Overview
 - Timeline
 - Parties
 - Documents
 - Evidence
 - Filings
 - Deadlines
 - Tasks
 - Notes
 - Approvals
 - Audit

Petta er aðal navigation model kerfisins.

11. Case Workspace

Case Workspace er mikilvægasti skjár kerfisins.

CASE-2026-00123	
Jón Jónsson v. Example ehf.	● OPEN

Overview Documents Evidence Filings Timeline...	
Case summary	
Next deadline	14 Sep 2026
Responsible	Jane Doe
Priority	HIGH
Confidentiality	RESTRICTED
Recent activity	Upcoming deadlines
Document uploaded	14 Sep – Court filing
Filing approved	21 Sep – Response
Evidence added	

12. Case Header

Case header skal alltaf sýna:

case number
case title
status
priority
confidentiality
responsible user
Optional:

case type
court
external reference

13. Confidentiality Indicator

Confidential cases skulu hafa mjög skýra visual indication.

Dæmi:

 RESTRICTED

eða equivalent design-system component.

Notandi á ekki að þurfa að opna metadata til að vita að case sé restricted.

14. Case Status Indicator

Status verður sýnilegur sem badge.

Dæmi:

DRAFT
OPEN
INVESTIGATION
LITIGATION
SETTLED
CLOSED
ARCHIVED

Color er ekki eina leiðin til að skilgreina status.

Texti + icon/state er required fyrir accessibility.

15. Dashboard

Dashboard sýnir það sem notandi þarf að vinna úr.

My Cases
Upcoming Deadlines
Pending Approvals
Recent Activity
Assigned Tasks
Notifications
Ekki gera dashboard að almennu analytics dashboardi í MVP.

16. Case List

Case list:

Case	Title	Status	Priority
2026-00123	Jón v Example	OPEN	HIGH
2026-00124	ABC contract	DRAFT	MEDIUM

Features:

search
filter
sort

pagination
saved filters

17. Case Search

Search fields:

case number
title
party
responsible user
status
type
date

Search results verða security-filtered server-side.

18. Document Browser

Document view:

Documents

- Pleadings
- Correspondence
- Contracts
- Evidence
- Court Documents
- Other

Document list:

Name
Type
Version
Uploaded by
Uploaded
Classification
Status

19. Document Viewer

Desktop layout:

document.pdf	Download ▼
--------------	------------

Document Preview	Metadata Version Classification Uploaded by Hash
---------------------	--

Viewer má ekki þurfa að senda binary document gegnum frontend server.

Signed/authorized URL er preferred.

20. Document Download

Flow:

User clicks Download

↓

Frontend → Backend

↓

Authorization

↓

Backend generates short-lived URL

↓

Browser downloads object

Frontend má ekki sjálfst búa til object-storage credentials.

21. Document Upload

Upload workflow:

Select file

↓

Metadata

↓

Classification

↓

Upload

↓

Processing

↓

Available

Large files skulu nota resumable/multipart upload þar sem við á.

22. Upload Progress

UI sýnir:

Uploading 72%
og síðan:

Scanning
Processing
Indexing
Available

Notandi skal ekki sjá document sem fully available áður en backend hefur staðfest processing state.

23. Evidence UI

Evidence list:

Evidence ID
Description
Type
Source
Linked document
Collected date
Custodian
Status

Evidence chain-of-custody upplýsingar skulu vera accessible úr UI.

24. Evidence Timeline

Dæmi:

01 Sep
Evidence collected
↓
02 Sep
Transferred
↓
03 Sep
Uploaded
↓
03 Sep
Hash verified

↓
04 Sep
Reviewed

25. Filing Workspace

Filing screen:

Filing
├ Details
├ Documents
├ Validation
├ Review
├ Approval
└ Submission

Workflow state alltaf visible.

26. Filing Workflow UI

DRAFT
↓
REVIEW
↓
APPROVED
↓
SUBMITTED
↓
ACCEPTED
Current state highlight-að.

Forbidden actions eru ekki aðeins hidden; þar sem við á skal UI útskýra hvers vegna action er unavailable.

27. Approval UI

Approval screen:

Approval requested

Requested by: John
Date: 01 Sep 2026

Documents:

filing.pdf
appendix.pdf

[Approve]

[Reject]

Approve action krefst confirmation.

28. Dangerous Actions

Actions sem geta haft legal consequences skulu hafa extra confirmation.

Dæmi:

Submit filing

Close case

Archive case

Delete document

Release document

Approve filing

Confirmation skal útskýra consequence.

29. Deadlines

Deadline dashboard:

Overdue

Today

Next 7 days

Next 30 days

Visual prioritization:

OVERDUE

URGENT

UPCOMING

NORMAL

30. Calendar

Calendar view er secondary.

Primary deadline management:

list

Calendar:

optional visualization

List view er betri fyrir actionable legal deadlines.

31. Tasks

Task:

Task

- title
- case
- assignee
- due date
- priority
- status
- related resource

Actions:

Complete

Reassign

Change due date

Open case

32. Global Search

Search UI:

Search cases, documents, parties... |

Results:

Cases

Documents

Evidence

Parties

Filings

33. Search Security

Critical rule:

Search má aldrei sýna resource sem notandi hefur ekki permission til að sjá.

Þetta á einnig við um:

result counts
facets
autocomplete
metadata
snippets

34. Audit UI

Audit page:

Timestamp
User
Action
Resource
Case
Result
IP
Dæmi:

14:31 John Doe
DOCUMENT_DOWNLOADED
CASE-2026-00123
SUCCESS
Audit content má ekki vera editable.

35. Timeline

Case Timeline sameinar significant events:

Case created
Party added
Document uploaded
Document reviewed
Deadline created
Filing approved
Filing submitted
Case status changed
Timeline er derived presentation, ekki authoritative event store.

36. Notifications

Notification types:

deadline approaching
task assigned
approval requested
filing status changed
document processing complete
system warning
Notifications skulu ekki innihalda viðkvæm gögn sem ekki þarf.

37. Authentication UX

Initial flow:

Open application
↓
Identity Provider
↓
Authentication
↓
Redirect
↓
Application
Frontend geymir ekki password.

38. Session Handling

Frontend skal nota secure authentication mechanism.

Preferred:

HttpOnly Secure SameSite cookies
ef backend architecture styður það.

Avoid:

long-lived access token
í localStorage.

39. Authorization in Frontend

Frontend getur notað permissions til að:

hide actions
disable actions

show navigation

En:

Frontend authorization er UX control, ekki security boundary.

Backend framkvæmir endanlega authorization.

40. Permission Hook

Example:

```
const canApprove = usePermission(
  "FILING:APPROVE"
);
```

UI:

```
{canApprove && <ApproveButton />}
```

41. Optimistic Updates

Ekki nota optimistic update fyrir legal state changes.

For example:

Approve filing

Close case

Submit filing

Archive case

skal:

request

↓

server confirms

↓

UI updates

42. Safe Optimistic Updates

Má nota optimistic updates fyrir low-risk UI state:

filter selection

local preferences

collapsed panels

ekki fyrir authoritative legal state.

43. Data Fetching

TanStack Query:

`server state`

Zustand:

`client state`

Ekki geyma allt server state í global Zustand store.

44. Query Cache

Example:

`cases`

`cases/{id}`

`cases/{id}/documents`

`cases/{id}/deadlines`

`cases/{id}/timeline`

Mutations invalid-a viðeigandi queries.

45. API Client

OpenAPI schema frá backend verður source of truth.

Recommended:

OpenAPI

↓

generated TypeScript client

↓

feature API wrappers

↓

React hooks

Þetta minnkar mismatch milli frontend og backend.

46. API Error Model

Expected:

```
{
  "error": {
    "code": "CASE_NOT_FOUND",
```

```
    "message": "Case could not be found",  
    "request_id": "..."  
  }  
}
```

Frontend notar `code`, ekki texta, til að ákvarða behavior.

47. Error UX

Notandi fær:

What happened

What can I do

Reference/request ID

Dæmi:

Unable to approve filing.

The filing may have changed since you opened it.

Request ID: 8e92...

48. Loading States

Ekki nota blank screen.

Use:

skeleton

spinner

progress

eftir context.

49. Empty States

Dæmi:

No documents

There are currently no documents associated
with this case.

[Upload document]

Empty state skal hjálpa notanda að taka næsta action.

50. Accessibility

Target:

WCAG 2.2 AA

Requirements:

keyboard navigation
screen reader support
focus management
semantic HTML
ARIA where required
sufficient contrast
non-color status indicators

51. Keyboard Navigation

Power users skulu geta unnið hratt.

Potential shortcuts:

/
global search

g c
cases

g d
documents

g t
tasks
Shortcuts skulu ekki trufla normal text input.

52. Internationalization

System architecture skal styðja:

Icelandic
English
Primary initial language:

Icelandic
UI strings mega ekki vera hard-coded inni í components.

53. Date & Number Formatting

Backend sends:

UTC / ISO 8601

Frontend formats according to:

user locale

Dæmi:

1 September 2026

eða íslensk locale formatting.

54. Legal Terminology

UI vocabulary skal vera centralized.

Dæmi:

Case

Document

Evidence

Filing

Deadline

Party

Approval

Translation glossary verður maintained centrally.

55. Responsive Design

Desktop:

3-column / 2-column layouts

Tablet:

2-column

Mobile:

single column

Full document review og case management er ekki optimized fyrst fyrir mobile.

56. State Management Rules

Local state

modal
form state
tabs
filters
UI toggles

Server state

cases
documents
deadlines
filings
→ TanStack Query.

Global client state

user preferences
navigation state
global UI state
→ Zustand.

57. Forms

React Hook Form + Zod.

Validation:

client-side
on backend validation:

authoritative

58. Form Error Handling

Validation errors:

field-level
Business errors:

form-level
Example:

Cannot submit filing:
approval is required.

59. Unsaved Changes

Ef user navigates frá formi með breytingar:

Unsaved changes

Your changes have not been saved.

[Leave]

[Stay]

Critical forms skulu styðja this behavior.

60. Document Editing

MVP:

download

preview

metadata

version

upload new version

Full browser-based document editing er ekki MVP requirement.

Possible future integration:

Microsoft 365

OnlyOffice

LibreOffice Online

61. Audit Visibility

User skal geta séð relevant history úr Case Workspace:

Timeline

Administrators geta séð full audit records:

Audit

Petta eru tvö mismunandi UX concepts.

62. Security UX

Security state skal vera visible.

Dæmi:

 Restricted

 Legal Hold

 Approved

 Processing

Notandi skal geta skilið state án þess að skoða raw metadata.

63. Legal Hold UX

Case með legal hold:

 LEGAL HOLD ACTIVE

Dangerous actions:

Delete

Purge

Destroy

eru disabled.

UI sýnir:

This item is protected by legal hold.

64. AI UX

AI features skulu vera greinilega merkt.

Dæmi:

AI-generated summary

Ekki:

Summary

AI-generated output skal sýna:

AI generated

Generated at

Model/provider where appropriate

Source documents

65. AI Review

AI output:

Generated summary

...

Source documents:

- Document A
- Document B
- Document C

[Accept]

[Edit]

[Reject]

Human review er required fyrir consequential outputs.

66. AI Hallucination Protection

AI UI skal hvetja til source verification.

Possible:

View source

við claims þar sem hægt er.

AI summary má ekki fela source provenance.

67. Frontend Security Controls

Frontend skal:

CSP

HTTPS only

secure cookies

dependency scanning

XSS protection

CSRF protection where applicable

Ekki treysta á frontend input sanitization sem security boundary.

68. Content Security Policy

Production CSP skal restrict-a:

script-src

connect-src

img-src

font-src

`frame-src`
eins mikið og practical.

Document viewer integrations skulu vera explicit allow-listed.

69. XSS Protection

Notandi eða document content má aldrei render-a arbitrary HTML án sanitization.

Sérstaklega:

`document metadata`
`AI output`
`notes`
`party names`
`external content`

70. Performance Targets

Initial targets:

`Initial application load`
`< 3 sec on normal corporate network`

`Case list`
`< 1 sec after cached API response`

`Case workspace`
`< 2 sec typical`

`Search`
`< 2 sec typical`
Large documents eru undanskilin normal page-load targets.

71. Large Lists

Frontend skal nota:

`pagination`
eða:

`virtualization`
fyrir mjög stór datasets.

Ekki load-a:

10,000 cases
í browser.

72. File Upload Limits

UI skal lesa limits frá backend configuration þar sem possible.

Ef file er of stór:

File exceeds maximum allowed size.
Ekki bíða eftir upload til að segja notanda þetta.

73. Browser Storage

Ekki vista:

documents
secrets
access tokens
sensitive case content
í localStorage.

Local storage má nota fyrir:

non-sensitive preferences

74. Frontend Observability

Frontend telemetry:

page load
API latency
JavaScript errors
failed mutations
document processing UX failures
PII/case data skal ekki fara sjálfkrafa í telemetry.

75. Frontend Logging

Production browser logs skulu vera minimal.

Ekki:

```
console.log(case)
console.log(document)
console.log(token)
```

76. Testing

Unit

```
hooks
utilities
permission logic
formatters
state machines
```

Component

```
forms
tables
dialogs
case header
workflow components
```

Integration

```
API + UI
authentication
authorization
workflow
```

E2E

```
login
create case
upload document
approve filing
search
close case
```

77. E2E Test Example

```
Login
↓
Open Cases
↓
Create Case
```

↓
Open Case Workspace
↓
Upload Document
↓
Wait for processing
↓
Create Filing
↓
Request Approval
↓
Login as Approver
↓
Approve
↓
Submit
Þetta verður mikilvægasta end-to-end legal workflow testið.

78. Visual Regression

Critical screens:

Login
Dashboard
Case Workspace
Document Viewer
Filing Workspace
Search
Audit
geta fengið visual regression tests.

79. Design System

Shared components:

Button
Input
Select
DatePicker
Table
Badge
Dialog
Drawer

Tabs
Toast
Alert
Tooltip
Timeline
FileUpload
DocumentPreview

80. Domain Components

Feature-specific:

CaseStatusBadge
ConfidentialityBadge
CaseHeader
DeadlineList
FilingStatus
ApprovalPanel
EvidenceTimeline
DocumentVersionList
Petta eru ekki generic UI components.

81. Component Rule

Generic:

```
<Button />
```

Domain:

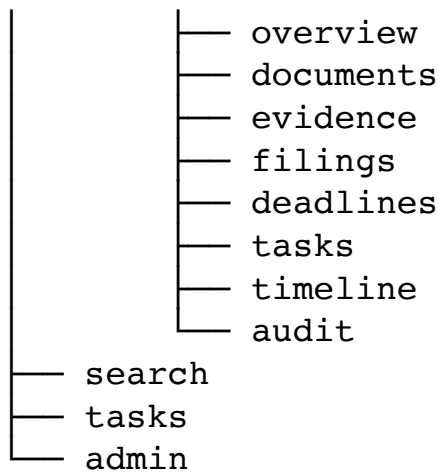
```
<ApproveFilingButton />
```

Domain component má innihalda business-aware UX logic en backend authorization er áfram authoritative.

82. Routing

Example:

```
/
├── dashboard
├── cases
│   ├── /cases
│   ├── /cases/new
│   └── /cases/:caseId
```



83. Route Guards

Route guards geta hindrað navigation.

Dæmi:

```
/admin  
ef user hefur ekki:
```

ADMIN:ACCESS

En backend þarf áfram að enforce-a permission.

84. Deep Links

Every resource page skal styðja direct URL.

Dæmi:

```
/cases/{caseId}/documents/{documentId}  
Notandi skal geta:
```

```
bookmark  
copy link  
open in new tab  
að því gefnu að permission sé til staðar.
```

85. Concurrency

Ef tveir notendur breyta sama resource:

Backend notar optimistic concurrency.

Frontend getur fengið:

409 Conflict

UI:

This item has been changed by another user.

[Reload]

Ekki overwrite-a silently.

86. Offline Mode

MVP:

No offline case editing.

Sensitive legal systems ættu ekki að reyna að styðja full offline editing nema sérstök architecture sé byggð fyrir það.

87. Accessibility + Security Combination

Critical actions skulu vera:

keyboard accessible

screen-reader accessible

visible

confirmation-based

Security controls mega ekki byggja eingöngu á:

color

hover

mouse interaction

88. UX Principle — Minimal Cognitive Load

Notandi á að geta svarað:

Where am I?

What case am I working on?

What is the current state?

What needs attention?

What can I do next?

innan nokkurra sekúndna.

89. UX Principle — Context Preservation

Pegar notandi opnar:

document
filing
evidence
deadline
innan case:

Case context skal varðveitast.

UI skal sýna breadcrumb:

Cases

/ 2026-00123
/ Documents
/ contract.pdf

90. UX Principle — No Silent Actions

Critical actions skulu alltaf hafa visible outcome.

Dæmi:

Submitting...
Submitted successfully.
Ekki:

button disappears
án explanation.

91. UX Principle — Explain Restrictions

Ekki bara:

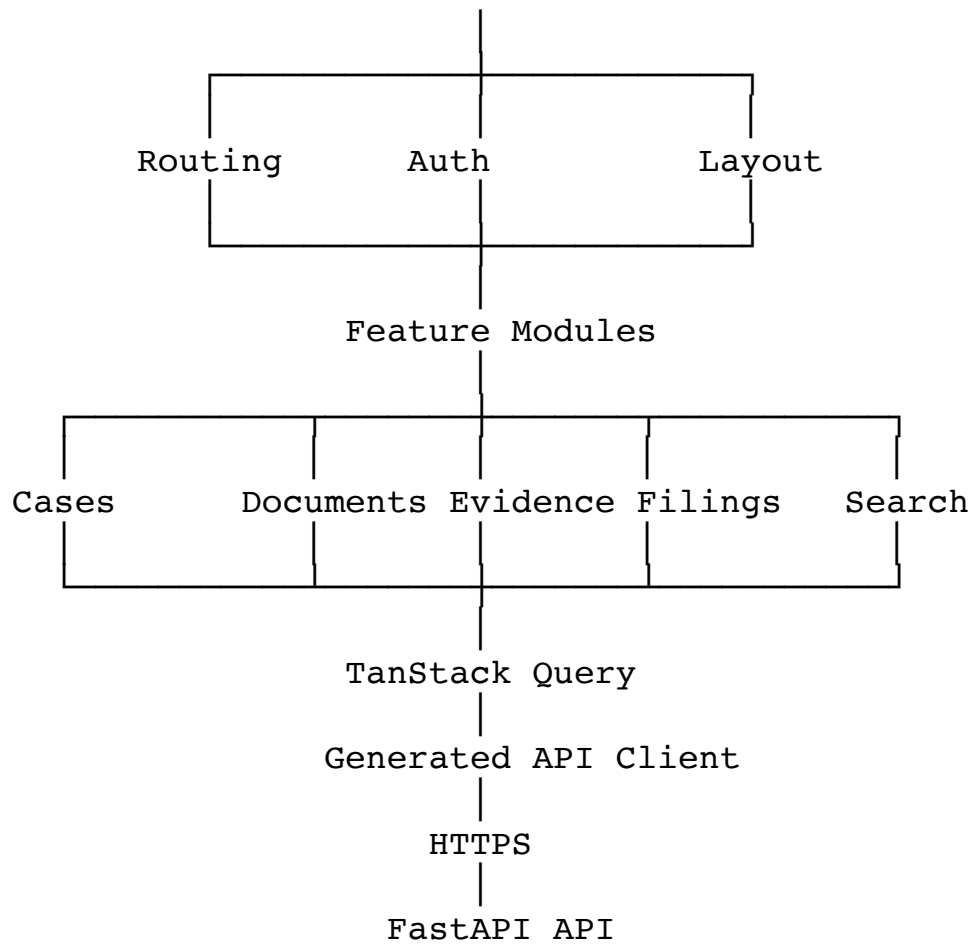
[Delete] disabled
Heldur:

Delete unavailable

This document is protected by an active legal hold.

92. Frontend Architecture Summary

React / TypeScript
|
Application Shell



93. Recommended First Frontend Vertical Slice

Byrja skal á:

Login



Dashboard



Case List



Case Workspace



Create Case



Edit Case



Timeline



Audit

Ekki byrja á öllum screens samtímis.

94. Frontend MVP Definition of Done

Frontend MVP telst tilbúið þegar:

- Authentication works
- Role-aware navigation works
- Case list works
- Case creation works
- Case workspace works
- Case status transitions work
- Documents can be listed
- Documents can be uploaded
- Document processing status visible
- Evidence can be viewed
- Filing workflow visible
- Approval workflow works
- Deadlines visible
- Tasks visible
- Search works
- Timeline works
- Audit works
- Confidentiality indicators work
- Legal hold indicators work
- AI-generated content clearly identified
- WCAG 2.2 AA baseline achieved
- E2E critical workflow passes
- Security tests pass

95. Architecture Decision — Case Workspace

Decision:

Case Workspace er primary working environment.

Users should spend most of their time inside:

`/cases/:caseId`

rather than navigating repeatedly through global modules.

96. Architecture Decision — Server State

Decision:

TanStack Query verður authoritative client-side mechanism fyrir server state.

Zustand verður aðeins notað fyrir client/UI state.

97. Architecture Decision — API Types

Decision:

TypeScript API types verða generated frá backend OpenAPI specification.

Handwritten duplicate API models skulu forðast.

98. Architecture Decision — Security

Decision:

Frontend permissions control UX.

Backend permissions control security.

Petta er explicit trust boundary.

99. Architecture Decision — Legal Actions

Decision:

Legal/consequential actions verða pessimistic.

`request`

→ `server validation`

→ `authorization`

- transaction
- response
- UI update

100. Architecture Decision — Mobile

Decision:

Desktop-first.

Mobile styður:

dashboard

notifications

deadlines

basic case information

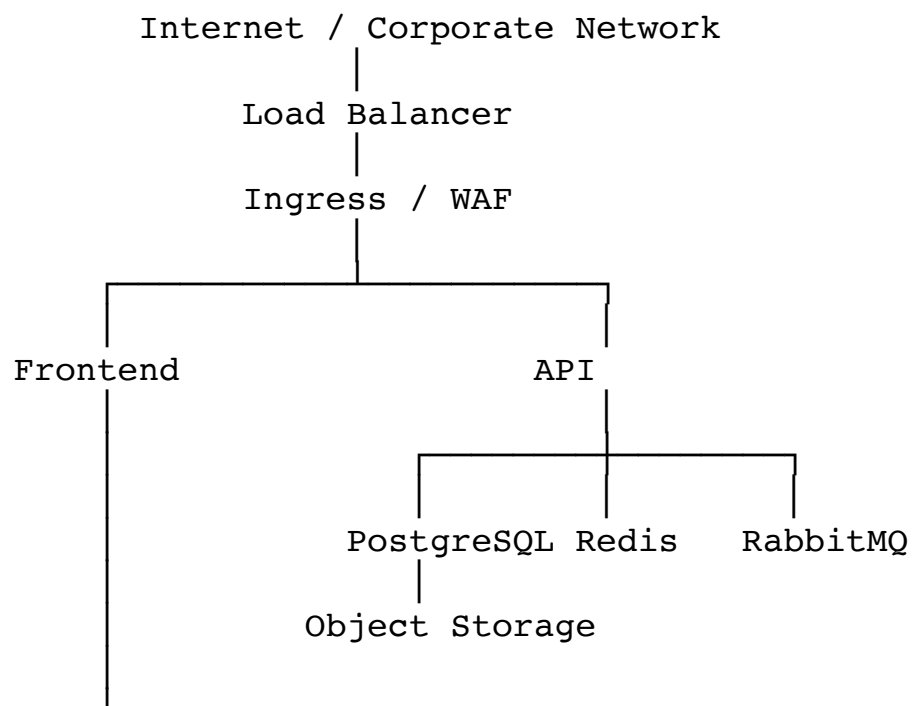
Full document/case workflow er desktop-first.

101. Next Specification — v1.0

Með v0.9 erum við komin með backend + frontend architecture.

Næsta skref ætti því að vera **Technical Specification v1.0 — Infrastructure, Deployment & Production Architecture**.

Það skjal myndi loka architecture-laginu með:



og skilgreina nákvæmlega:

- Docker images
- Kubernetes namespaces
- Deployments
- Services
- Ingress
- TLS
- PostgreSQL HA
- Object storage
- RabbitMQ
- Redis
- Secrets/Vault
- Backup & restore
- Disaster recovery
- Monitoring
- Logging
- OpenTelemetry
- CI/CD
- Git strategy
- environments
- staging/production separation
- network segmentation
- firewall policy
- container security
- vulnerability scanning
- SBOM
- database migration strategy
- zero/low-downtime deployment
- RPO/RTO

- disaster recovery testing
- production sizing

Þá verður architecture specification-ið í raun komið að þeim punkti að hægt sé að setja upp **fyrsta fulla development/staging environmentið** og hefja implementation.

Næsta eðlilega skref er því v1.0: að taka þessa hönnun úr application layer og yfir í **Infrastructure, Kubernetes, CI/CD, backup/DR og production security**. Það er líka rétti tímapunkturinn til að taka endanlega ákvörðun um hvort ILCMS eigi að miða við **on-premises, private cloud eða hybrid deployment**.